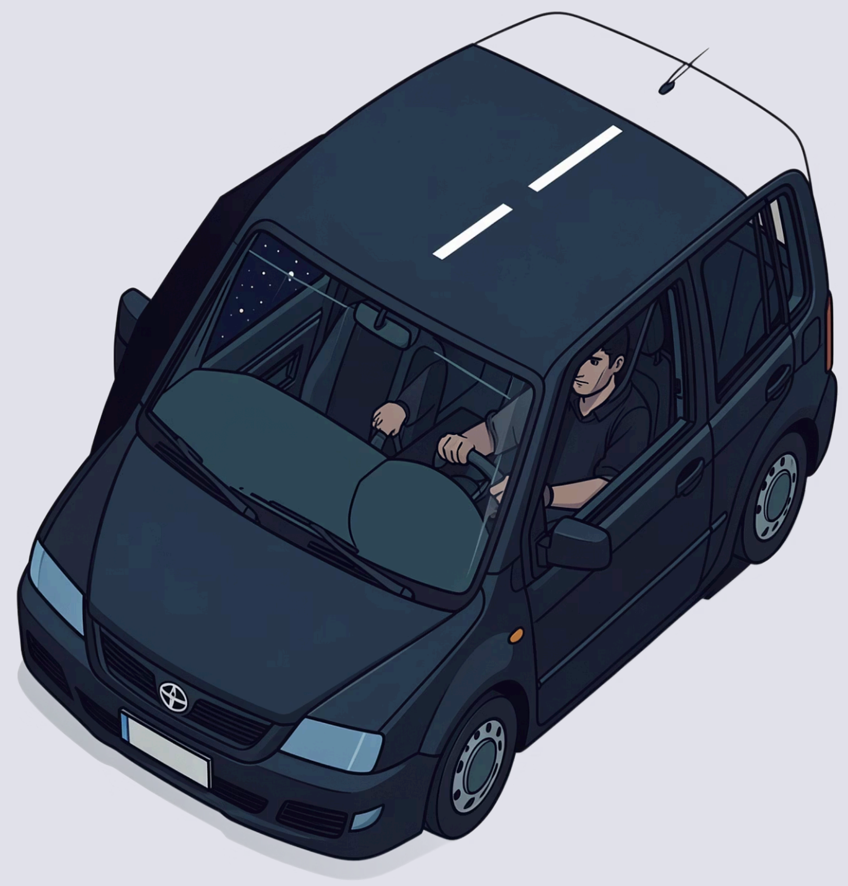
Driver Drowsiness and Distraction Detection System

Real-time facial monitoring to improve road safety

Student: Sasidhar reddy , santosh . **Faculty Guide:**[Lavanya madam]



Problem Statement

A Silent Danger

Drowsy driving is a major contributor to road accidents worldwide

Unnoticed Until Too Late

Fatigue-related lapses — closing eyes, yawning, losing focus — go undetected

High-Risk Scenarios

Long drives, night shifts, monotonous highways

No Self-Monitoring

Drivers cannot reliably detect their own drowsiness in real time

Motivation & Objectives

Motivation

An always-alert co-pilot — detecting early signs of drowsiness and distraction before a lapse turns into an accident.

Objectives

01

Detect facial landmarks (eyes, mouth) in real time from live camera feed

02

Classify driver state as **alert**, **drowsy**, or **yawning**

03

Trigger immediate audio-visual alert when drowsiness is detected

04

Evaluate robustness across lighting conditions and with/without glasses

Dataset

Yawn-Eye Dataset

Kaggle

4 classes: Open, Closed, Yawn, No Yawn

Trains eye-state and yawn classifiers

MRL Eye Dataset

84,898 images

Open/closed eye images under varying illumination

Strengthens eye-state classifier

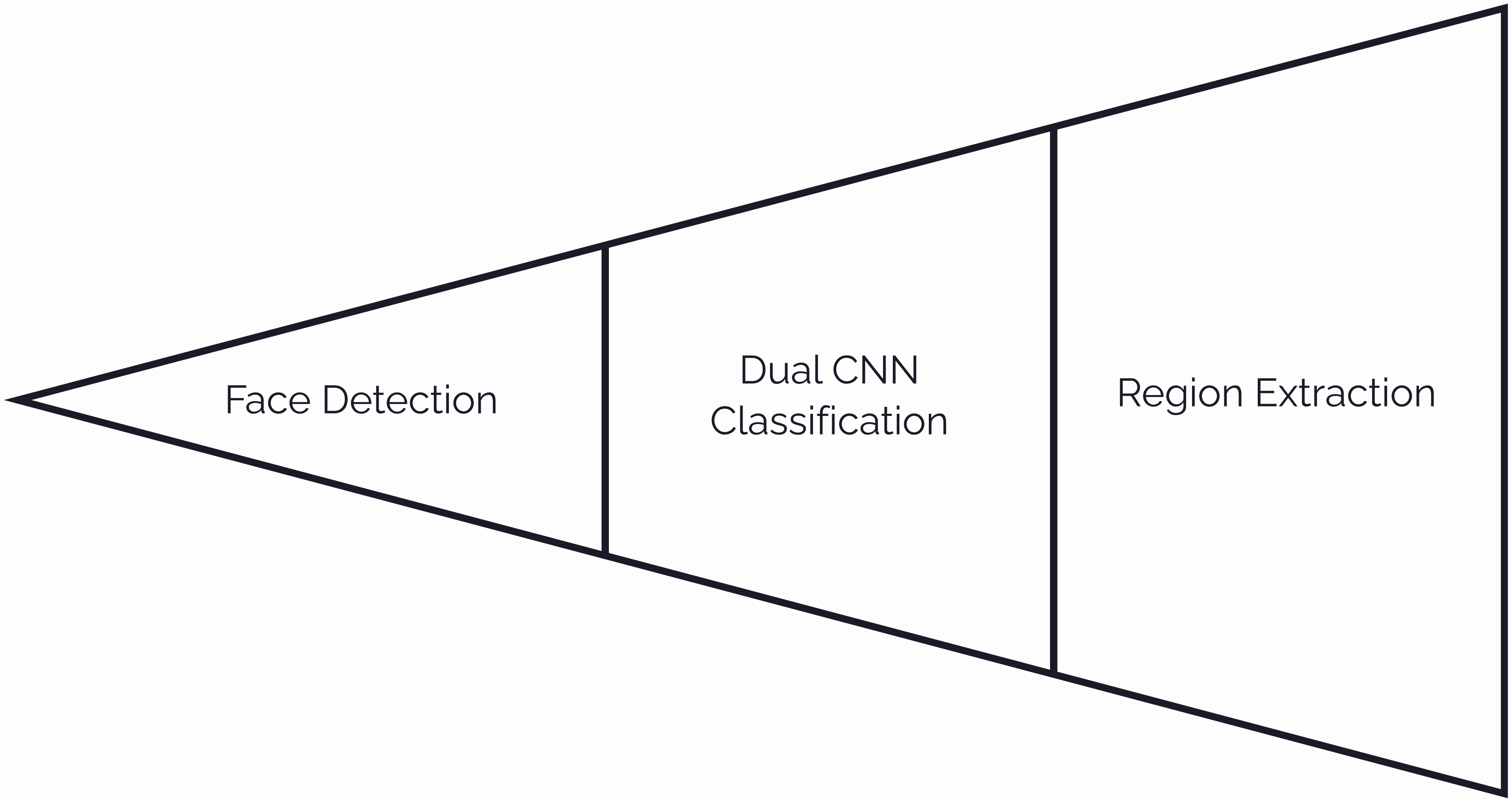
NTHU-DDD

66,521 images · 36 subjects

With/without glasses, day and night conditions

Tests generalization to realistic driving scenarios

Proposed Methodology



Face Detection & Landmarks

MediaPipe Face Mesh for real-time facial landmark localization

Classification

MobileNetV2 transfer learning for eye state and yawn detection

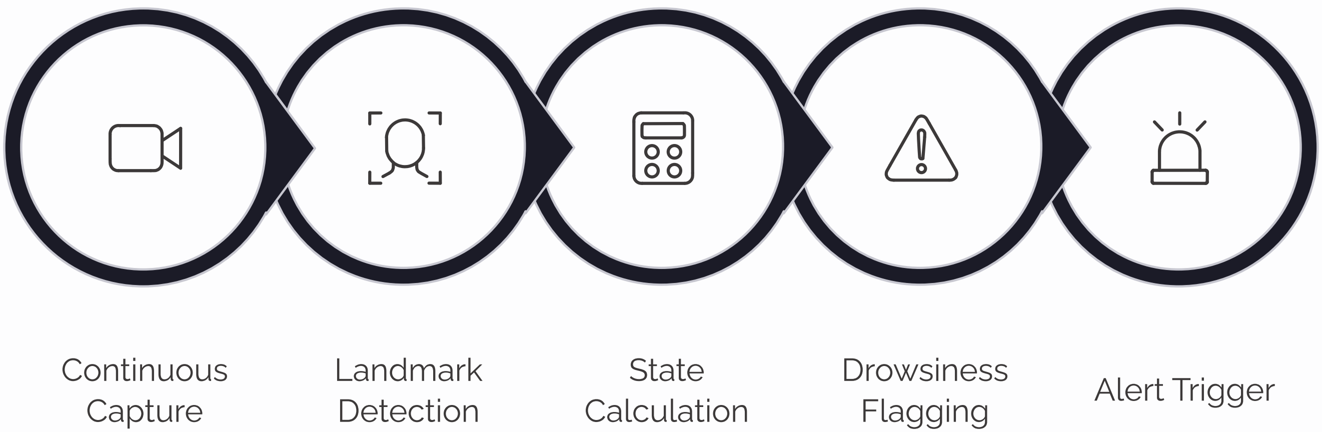
Hybrid Approach

Geometric analysis (EAR, MAR) combined with deep learning for robustness

Temporal Logic

Tracks duration and frequency — not single-frame decisions

Real-Time Detection & Alert Workflow



System Architecture

→ **Camera Input** — continuous driver face capture

→ **Processing Pipeline** — OpenCV + MediaPipe + CNN

→ **Alert System** — audio alarm + on-screen visual warning

⚠ Drowsiness flagged when eyes remain closed beyond safe threshold or yawning repeats within a short time window.

Robustness & Evaluation Approach

Lighting Conditions

Performance tested in both day and night scenarios

Eyewear Variations

Tested with and without glasses or sunglasses

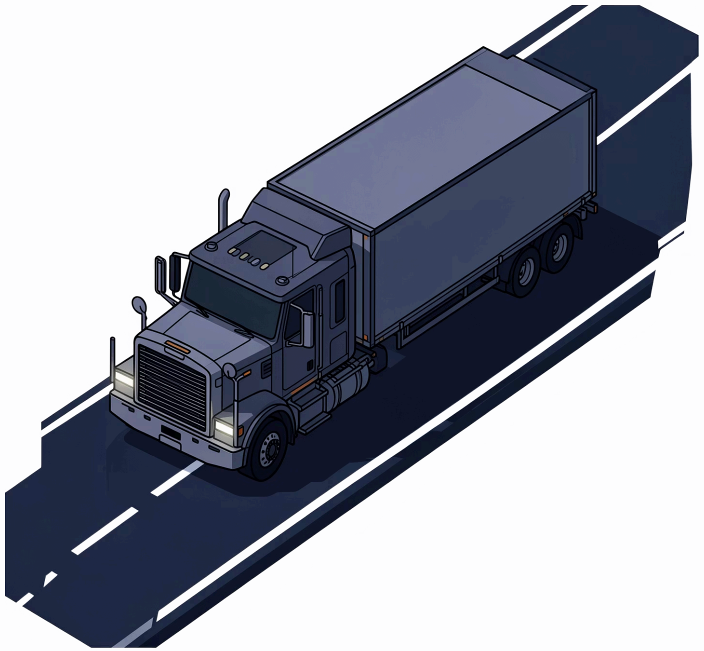
Demographic Diversity

Evaluated across different face shapes and skin tones

Evaluation Metrics

Accuracy, precision, recall, and false-positive rate under varied conditions

Real-World Usefulness & Impact



Who Benefits

- Long-haul truck drivers and commercial vehicle operators
- Night-shift workers and highway commuters
- Fleet operators seeking to reduce accident liability

Proactive Safety	Easy Integration	Non-Intrusive
Catches drowsiness before it becomes dangerous	Works with existing dashboard cameras or as low-cost aftermarket device	Passive background monitoring — no driver action required

Tools, Technologies & Project Timeline

Technology Stack

Languages & Frameworks

Python · OpenCV · MediaPipe · TensorFlow / PyTorch

Model Architecture

MobileNetV2 — lightweight CNN for real-time classification

Demo & Deployment

Streamlit + streamlit-webrtc for browser-based demo

Project Timeline

1 — Phase 1

Setup, landmark detection, EAR/MAR baseline

2 — Phase 2

CNN classifier training for eye and yawn states

3 — Phase 3

Real-time integration, temporal logic, alert system

4 — Phase 4

Robustness testing, demo packaging, final evaluation

Conclusion & Future Scope

Expected Contribution

- Real-time, non-intrusive driver monitoring detecting drowsiness and yawning
- Alerts driver before an accident occurs — improving road safety

Future Scope

→ Extend to detect phone usage and gaze deviation

→ Integrate with vehicle systems — auto-slow-down, emergency alerts

→ Deploy on low-cost embedded hardware for real vehicles

→ Expand to multi-occupant monitoring for passenger safety